ChurnGuide in Easy Words

Study guide for the B.Tech project presentation. Written from the actual code and the saved results in the project folder.

1. What is ChurnGuide?

ChurnGuide is a small web app that runs on your own computer. You give it a table of customers (a CSV file). It learns from customers who already left, and then gives every customer a **churn probability**: a number from 0% to 100% showing how likely that customer is to leave. It can also group customers into segments and show which inputs push the probability up or down.

Churn simply means a customer stops using a service or cancels it. Losing customers costs recurring income, and keeping a customer is usually cheaper than finding a new one, so an early warning helps a team act in time.

2. The data we used

- **Telecom (real public sample):** the public IBM Telco Customer Churn sample. 7,043 customers, 1,869 of them churned (26.5%). Columns include tenure (months), contract type, monthly charges, internet service and payment method.
- **B2B SaaS (synthetic):** 1,200 software accounts that were **generated by a script** in this project (scripts/generate_saas_data.py). It exists only to show that the app can accept a different set of columns. It is **not real customer data**, and its results are not proof that the method works for real software businesses.

3. How the app works, step by step

- **Step 1 - Add data.** Upload a CSV and give the project a name.
- **Step 2 - Check columns.** The app guesses the outcome column (for example Churn), what value means "left" (for example Yes) and the ID column. You can change these. Always check them yourself for a new file.
- **Step 3 - Train.** The data is split 80% for training and 20% for a final test (the split keeps the same churn share in both, with a fixed seed 42). Three models compete on the training part using 5-fold cross-validation: Logistic Regression, Random Forest and Gradient Boosting. The one with the best cross-validated ROC-AUC is chosen, without looking at the test part.
- **Results.** The chosen model is scored once on the 20% held-out test customers. Those are the numbers we report.

4. Results in plain words (telecom project)

Measure	Value	What it means
Accuracy	0.7984	About 80 of 100 test customers were classified correctly
ROC-AUC	0.8465	How well the model ranks leavers above stayers (0.5 = guessing, 1.0 = perfect)
Precision	0.6585	Of customers flagged as churners, about 66% really churned
Recall	0.50	The model finds only half of the real churners
F1	0.5684	One balance number of precision and recall

Test set: 1,409 customers. Counts: 938 correctly kept, 97 wrongly flagged, 187 churners missed, 187 churners found. The winner was Gradient Boosting (cross-validated ROC-AUC 0.8493, against 0.8463 for Logistic Regression and 0.8446 for Random Forest). **Note:** the 0.8493 on the app home page is this cross-validated ROC-AUC. The home card calls it "Top Accuracy", which is a wrong label.

Honest reading: the model is useful for ranking who is more at risk, but because recall is 50%, it misses half of the real churners. Whether a lower cut-off would help is something a business must test; we have not shown that.

5. The app pages

Page	What you do there
Home + simulator	Move contract, tenure and charge controls to see a live churn probability gauge.
ML Studio	The three-step wizard above: upload, check columns, train, see results.
Check one customer	Fill one customer profile (or use sample profiles) and get a probability, a risk tier and the top drivers.
Bulk scoring	Upload a file of customers without a churn column, get a probability and risk level for each, and download the scored CSV.
Analytics	Churn rate for each category of each column in the dataset.
Segments	Four customer groups made by K-Means clustering on tenure, monthly charges and total charges.
Diagnostics	Model comparison, held-out metrics, confusion matrix, ROC curve and permutation feature importance.
Project menu	Switch the active business (telecom or the synthetic SaaS project, or one you trained).

6. Risk levels and cut-offs

For one customer and the simulator, a probability of 0.60 or higher is labelled high risk. In bulk scoring the high-risk cut-off is 0.65. These are fixed numbers in the code, not tuned for any business. A sample "at-risk" telecom profile gives about 47.6% (Medium), so a profile named "at-risk" is not always labelled high.

7. Segments (telecom)

Segment (by typical values)	Customers	Historical churn
Low tenure, higher charges (median tenure 14 months)	2,276 (32.3%)	48.2%
Low tenure, low charges	1,704 (24.2%)	24.6%
High tenure, high charges	1,904 (27.0%)	15.4%
High tenure, low charges ("loyal core")	1,159 (16.5%)	5.0%

These churn rates are what happened in the dataset. They are not predictions. The suggested actions on the page are general ideas, not tested results.

8. How "why" explanations work

For one customer, the app changes one input at a time to a typical customer value (a median or most common value from the training data) and sees how the probability moves. This shows **model sensitivity**. It is not SHAP and it does not prove cause and effect. Missing baselines are filled with saved typical values, and TotalCharges is recalculated when tenure or charges change, so the effect is not strictly one-at-a-time.

9. Limits you should know

- Uses one snapshot per customer. There is no time-series or real-time monitoring.
- Recall is 50% on the telecom test set.
- The SaaS project is synthetic (accuracy 0.6542, ROC-AUC 0.6894, recall 0.49 on its test split). It is a software check, not validation.
- For a new CSV, the automatic guess of the outcome column can be wrong. Confirm it in the wizard (ML Studio) and do not rely on the quick "add business" shortcut for unfamiliar data.
- The "Download Scored CSV" file is a single shared file. Score the batch again after switching business, then download.
- Some home page cards ("Zero Leakage", "<10 ms", "4 Personas") are interface wording that we did not verify. The leakage controls are real design steps (pipeline fitted on training data only) but they do not prove the absence of leakage in new data.
- The home page overflows on a phone screen. Use a desktop browser.
- The Windows/Mac launchers (START_HERE files) were not tested by us. Tests with Python 3.11 gave 17 passed, 1 failed (an old navigation test that expects a redirect; the home page now returns normally).

10. How to run it

If you already have a working install, leave it as it is. Do not reinstall anything. Just start the app the way you already do.

For a fresh computer (recommended path): use Python 3.11 and a separate virtual environment inside the project folder, so nothing on the computer is changed. Open a terminal in `01_Project/ChurnGuide_BTech_Project` and run:

Windows:

```
py -3.11 -m venv .venv
.venv\Scripts\activate
python -m pip install -r requirements.txt
python app.py
```

Mac / Linux:

```
python3.11 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
python app.py
```

The app opens in your browser at the local address shown in the terminal. If Python 3.11 is not installed, install it first from python.org. Do not run pip outside the virtual environment.

What we tested: the app, the training and the tests were run by us with Python 3.11 in an isolated environment on Linux. The Windows and Mac commands above are the standard virtual-environment steps and were not run by us. The START_HERE.bat / .command / .sh launchers came with the project and were **not tested** by us, so they are not the recommended path.

11. Small glossary

Term	Meaning
Cross-validation	Splitting the training data into 5 parts, training on 4 and checking on 1, five times.
Held-out test set	20% of customers never used for training or choosing the model.
ROC-AUC	A score for ranking quality, from 0.5 (guessing) to 1.0.
Leakage	Using information that would not be known at prediction time, which makes results look too good.
K-Means	A method that groups similar customers without using the churn label.
Pipeline	Cleaning, scaling and the model chained together so each step is fitted on training data only.